

Samsung PIM/PNM for Transformer based AI

: Energy Efficiency on PIM/PNM Cluster

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Memory Trends and Bottleneck in HPC/AI

02

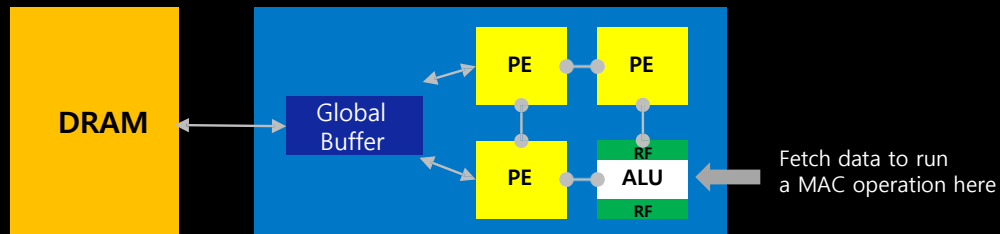
Samsung Memory AI Solution

- Cache Level LLC DRAM, **HBM-PIM (Processing-in-Memory)**
- Memory Expander, **CXL-PNM (Processing-near-Memory)**
, CXL-SSD
- Storage : SmartSSD, PBSSD

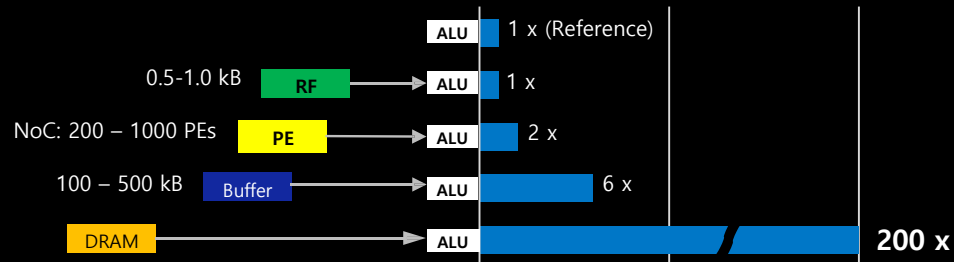
PIM/PNM on Memory Hierarchy and Energy Reduction

- **Data movement** consumes a lot of energy even for simple computation
- **PIM/PNM** technology can reduce energy consumption within a typical memory hierarchy
- PIM/PNM device for each layer must meet **specific requirements**: bandwidth(BW), power, capacity, etc.

Energy Cost of Data Transfer

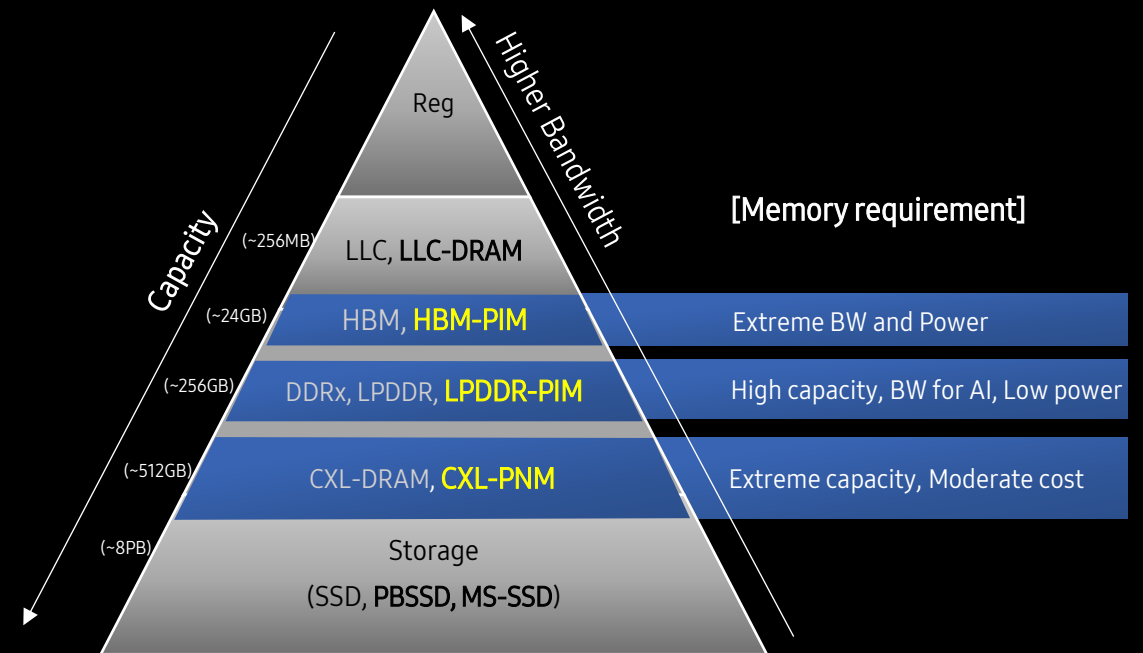


Normalized Energy Cost*



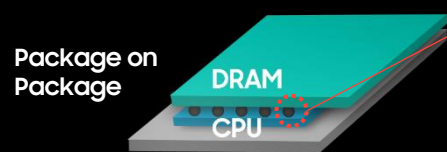
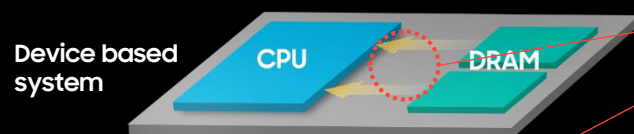
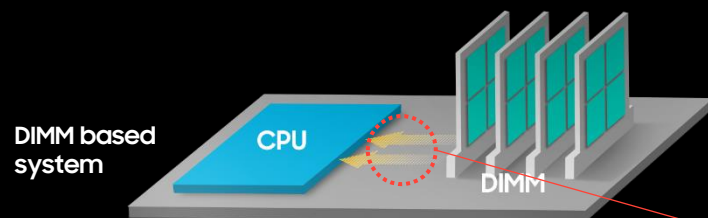
Source: * Y.-H. Eyeriss, 2016 ISCA

Computing Coupled Memory Hierarchy

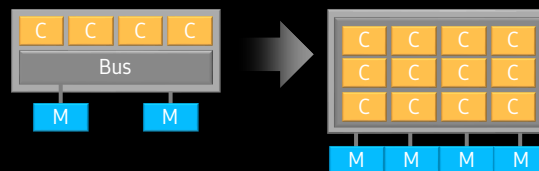


Traditional Approach to Overcome Memory Bottleneck

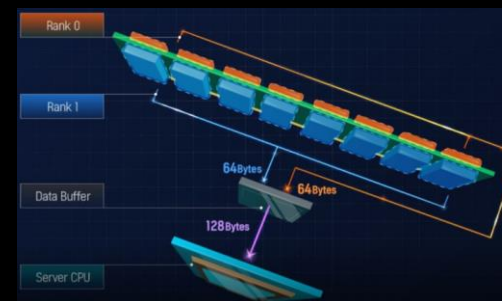
- While there are various methods to **increase BW**, it is difficult to achieve a dramatic increase
 - Limited by # of PCB wires, # of CPU ball, and thermal constraints
- Increasing # of the balls and PCB wires is **physically and thermally** bounded and is a expensive solution
 - MCR-DIMM, PAM3/4 signaling IO, 2K-IO or 3D stacking



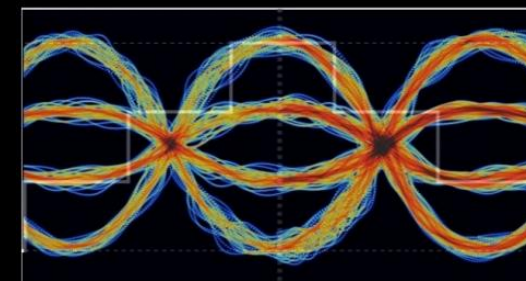
Memory BW is continuously requested due to AI application and multi-core host



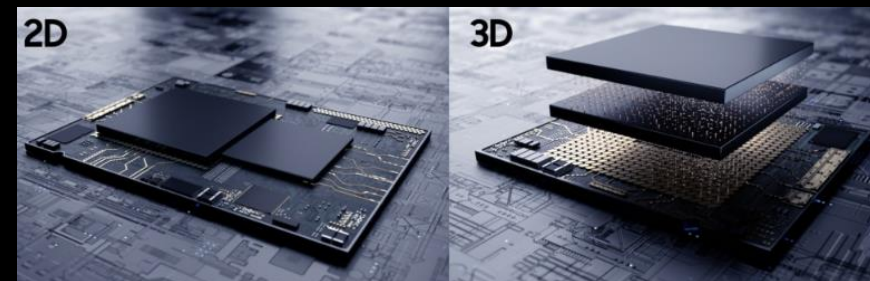
MCR-DIMM(Multiplexer Combined Ranks DIMM)



PAM 3/4(Pulse Amplitude Modulation)



2K-IO 2D or 3D stacking

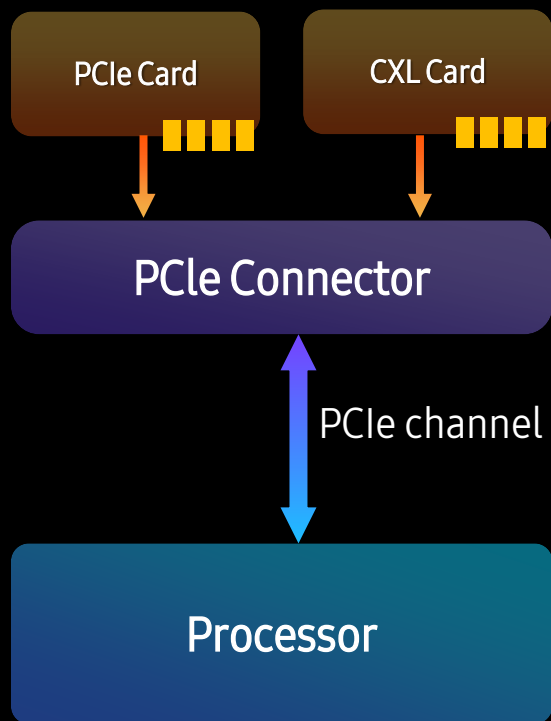


[source] Samsung

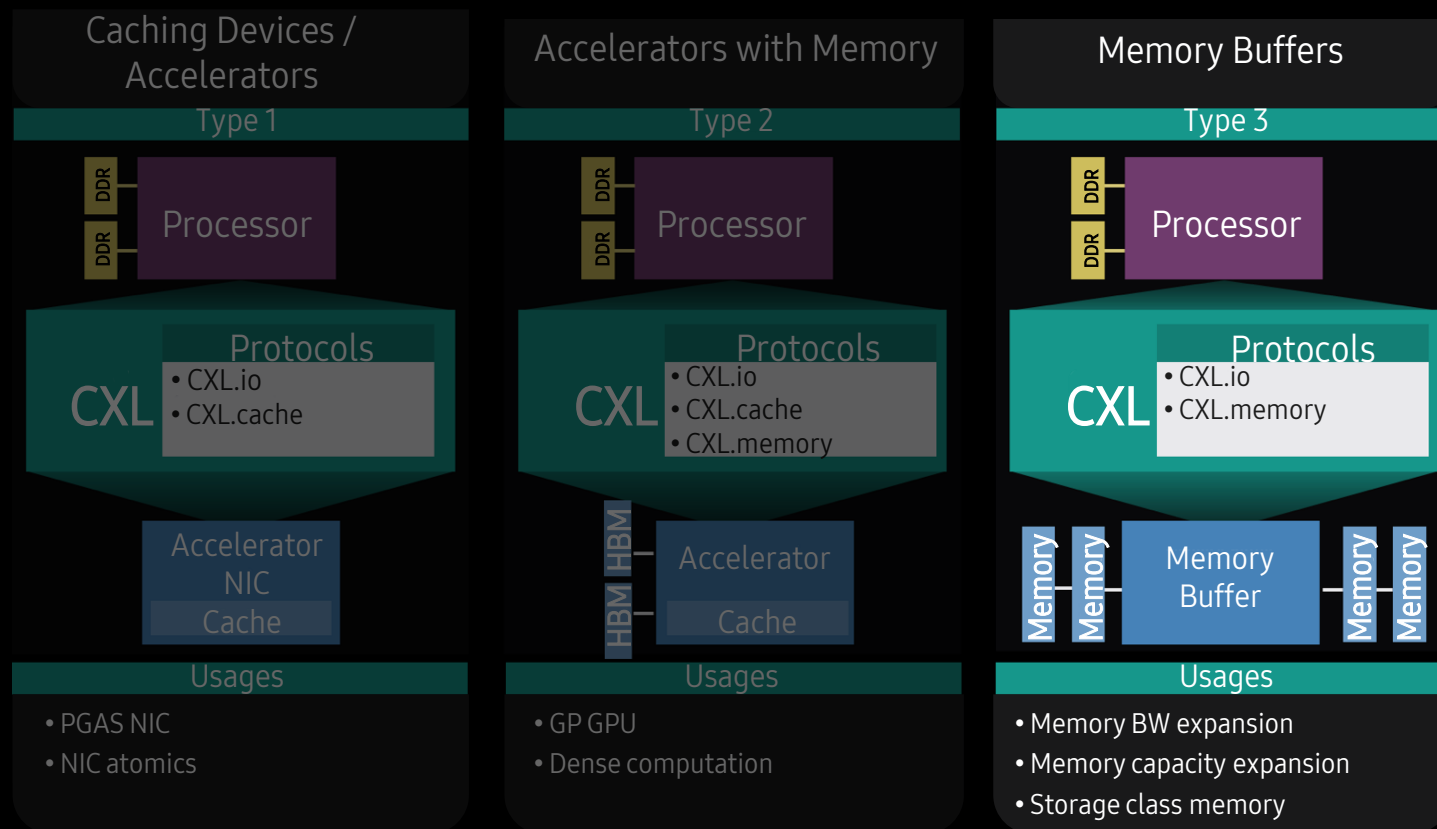
CXL solution, Trend to Pay Attention to

- CXL is strong candidate for memory hierarchy to address performance and density
- Successful power-on of **memory expander, SSD/pooling solutions** are next big-thing

CXL is a high performance, low latency protocol that leverages PCIe physical layer



CXL is an open industry standard with broad industry support



Bottleneck in GPT: Linear Layers in Generation Stage

- Main target: transformer **decoders** used in ChatGPT, GPT-3
 - Linear layers in multi-head attention(**MHA**) and feed-forward networks(**FFN**)
- Focus on memory-bottleneck in Generation stage
 - Generation Stage shows poor performance with GPU due to its memory-bound & sequential characteristic

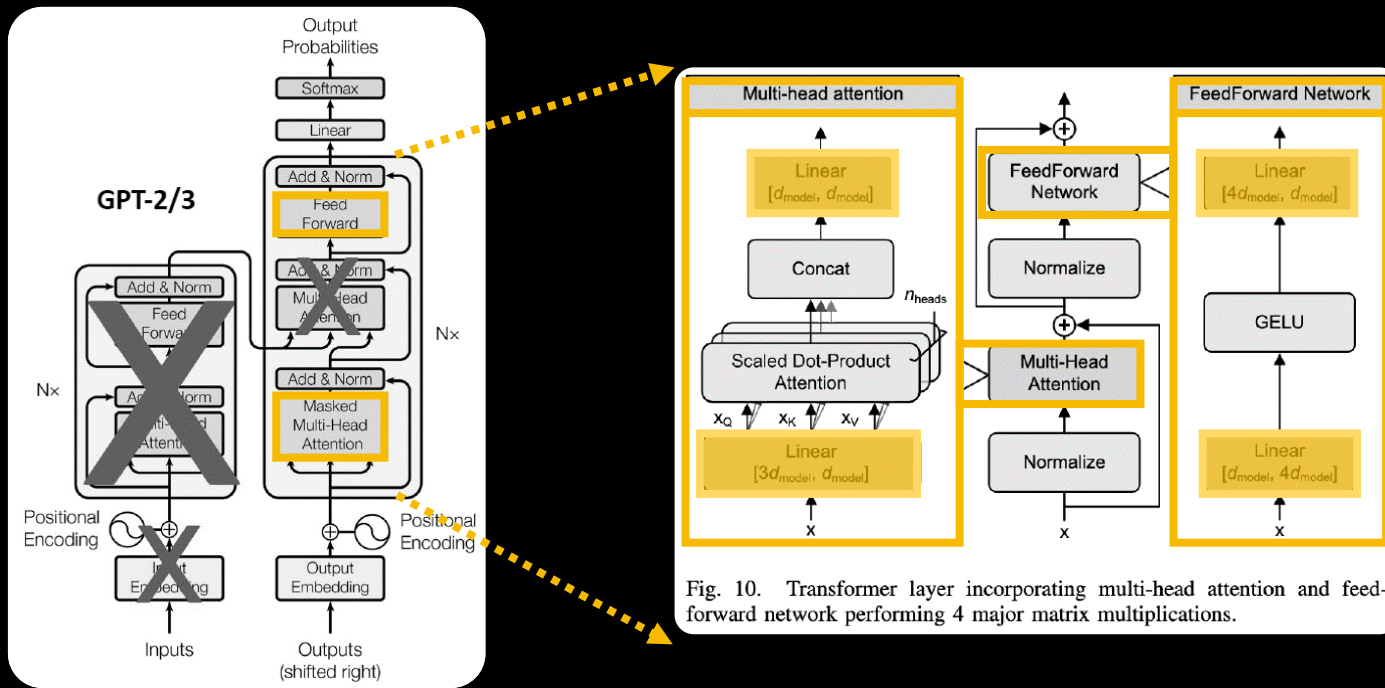
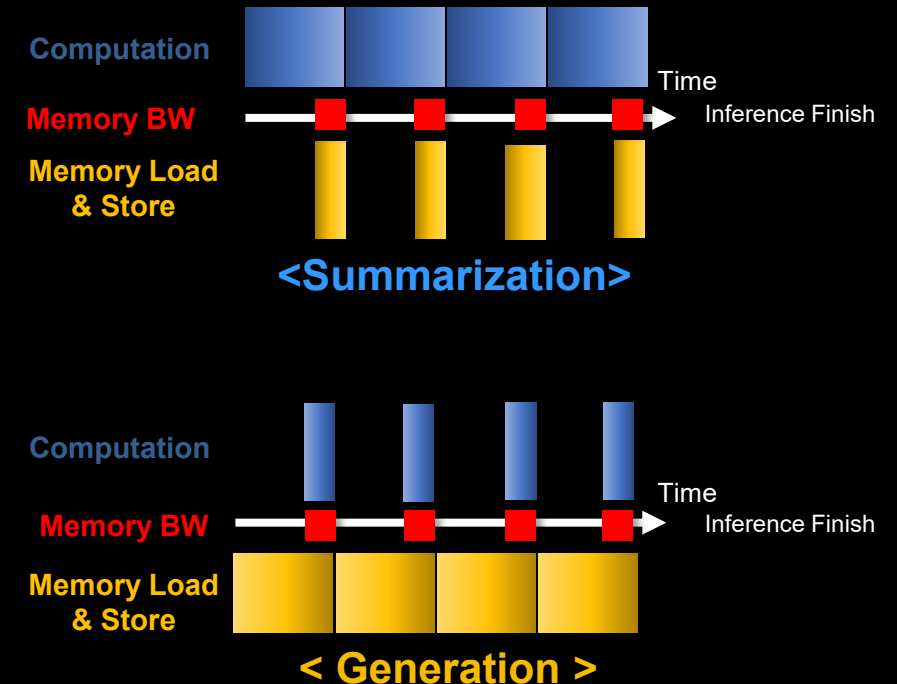


Fig. 10. Transformer layer incorporating multi-head attention and feed-forward network performing 4 major matrix multiplications.



<Transformer>

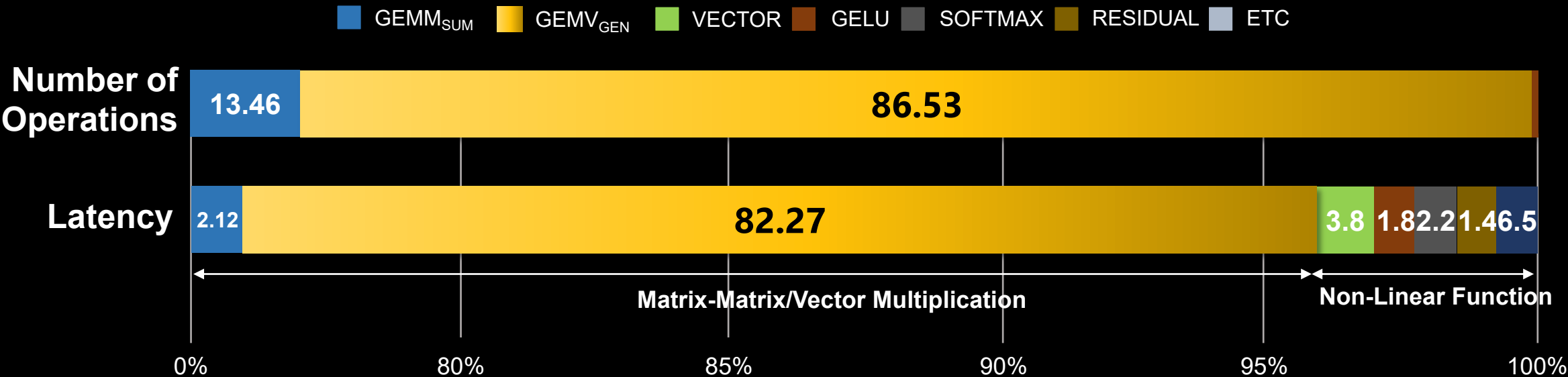
<Four major matrix multiplication block>

[source] Google Transformer

[source] Naver Clova

GPT Profiling Result

- GPT workload consists of Summarization(computing-bound) and **Generation(memory-bound)**
- GEMV portion can be **60–80%** of total generation latency, which are **the target of PIM/PNM**



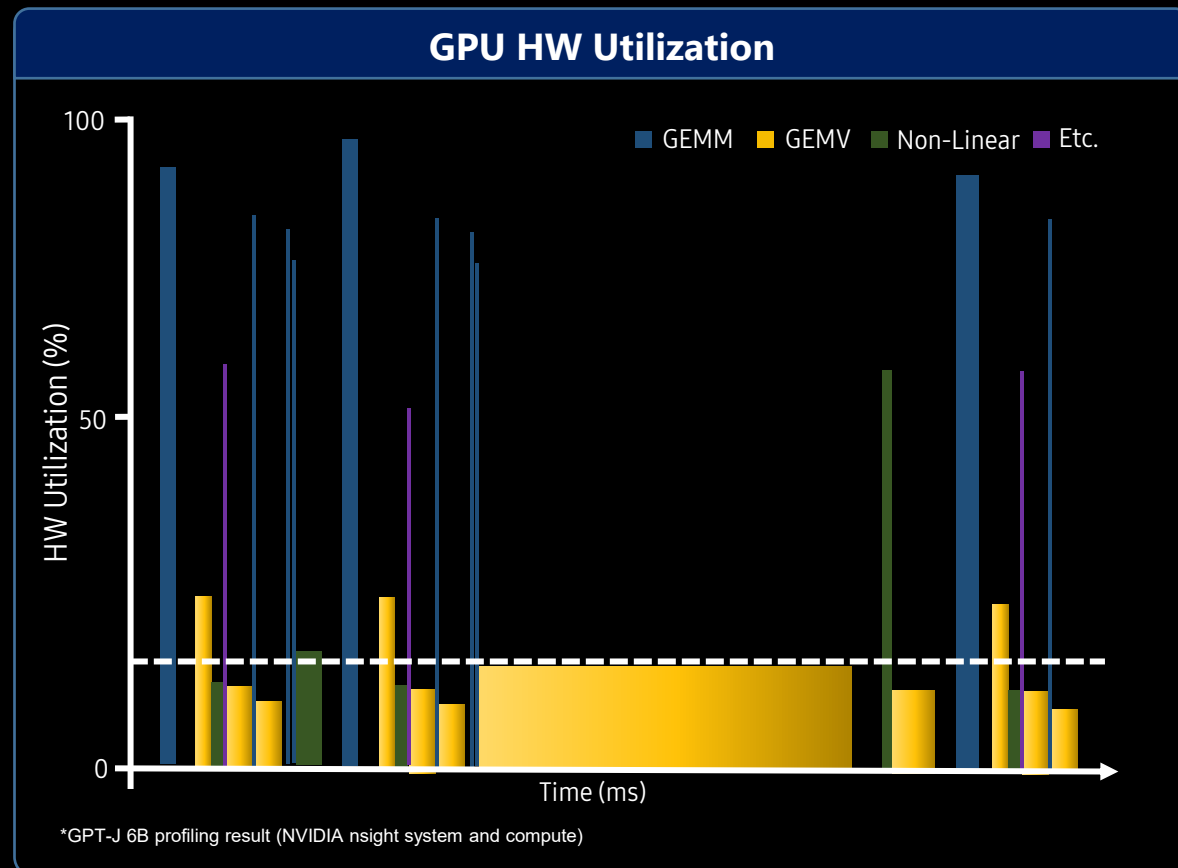
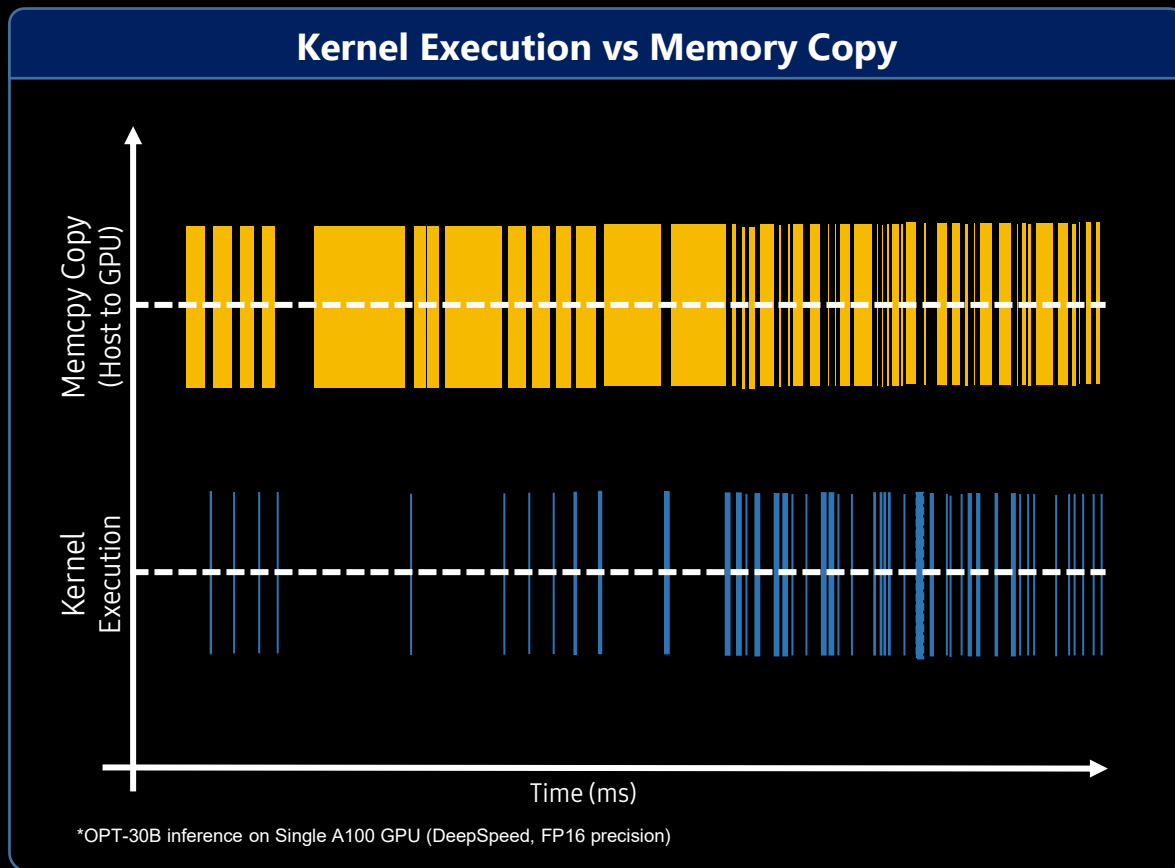
*Profiling result is measured in A100 System (DeepSpeed + GPT-J 6B, FP16, Input/Output token:7/46)
GPT-j: Google JAX framework

<https://community.openai.com/t/how-does-chatgpt-have-such-massive-token-limit/25738/6>

Stage	Computation	Latency
SUM	78.95 GFLOP	7.62 ms
GEN	11.28 GFLOP	6.58 ms

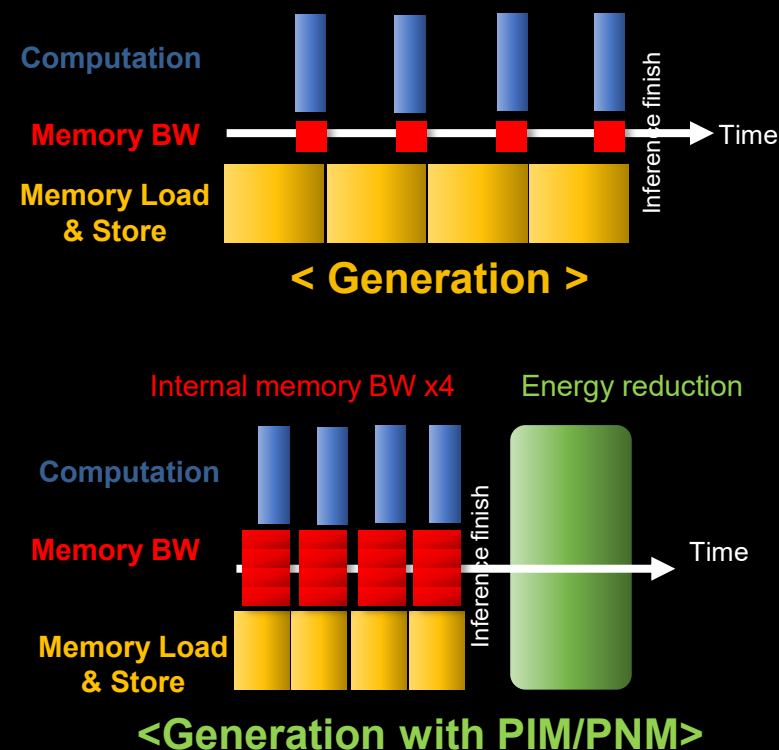
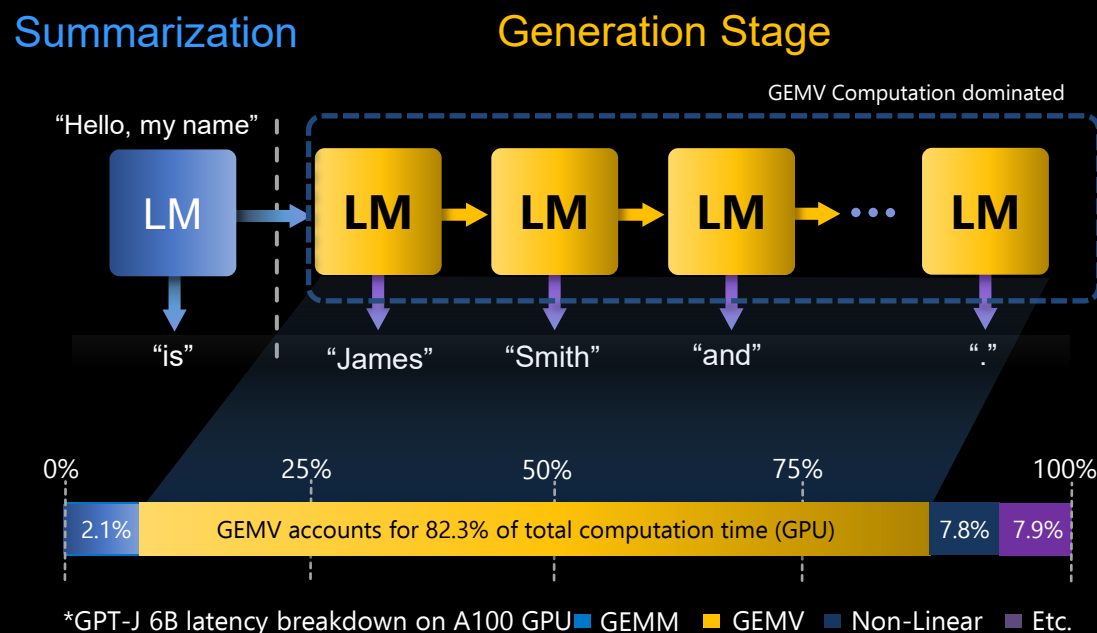
Utilization and Execution Time Breakdown

- Most of the execution time is spent for the memory copy from the host CPU memory to the GPU memory
- **Utilization** for performing **GEMV** operations (Generation stage) is **seriously low**, compared to GEMM
- As # of output tokens increases, GEMV operations dominate the inference time



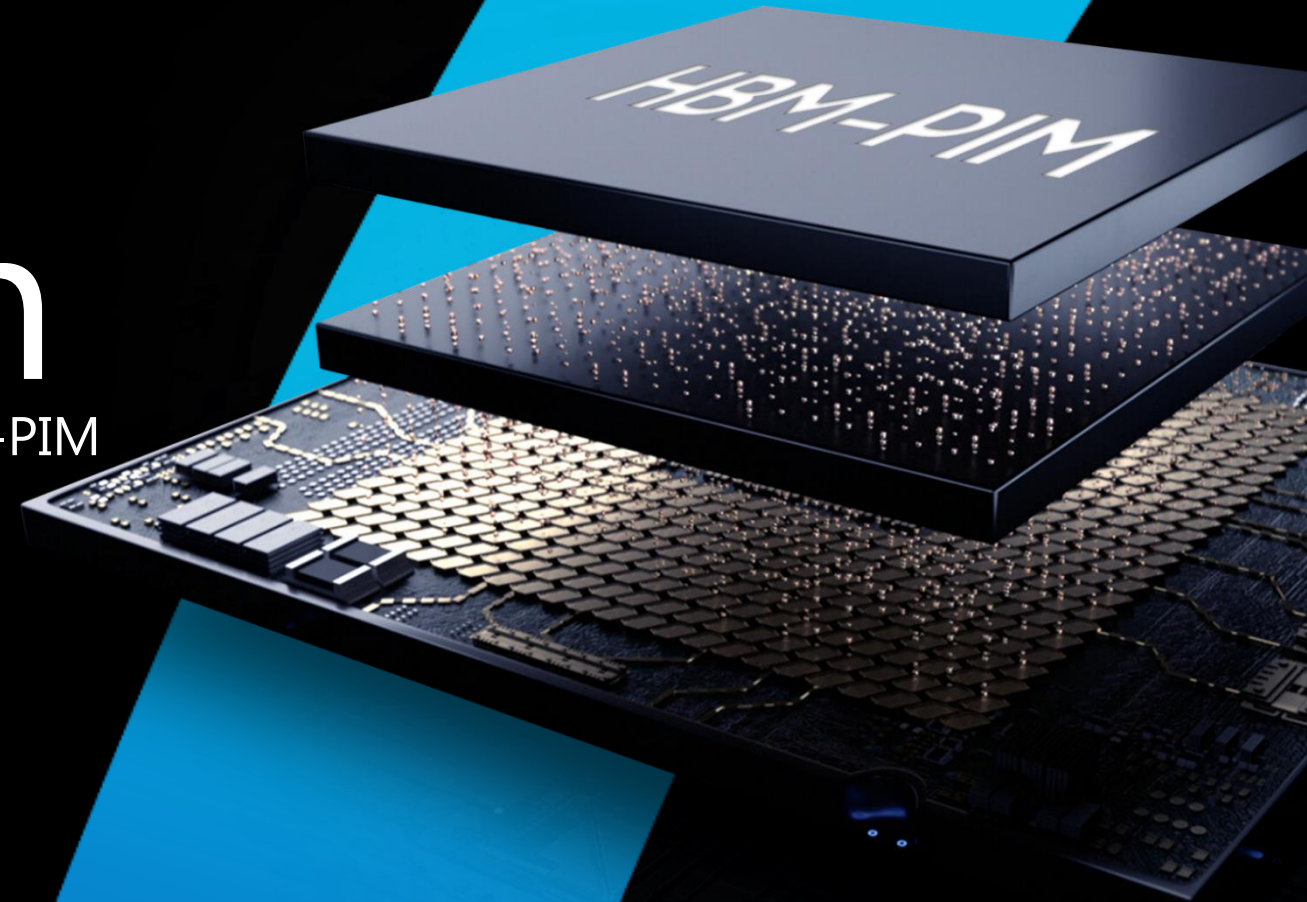
Acceleration by PIM/PNM on Generation stage

- Generation stage on GPT requires high capacity and bandwidth memory
- **MHA and FFN** can be fully offloaded to PIM/PNM, exploiting full bandwidth provided by PIM/PNM
- As a result, PIM/PNM can significantly reduce **the time and energy** spent on inference



PIM Solution

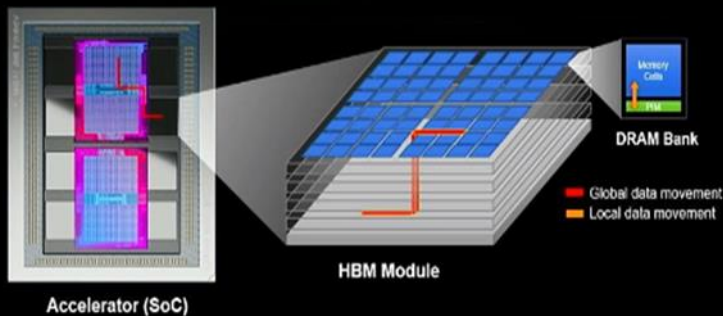
Redesigned to Advance AI : HBM-PIM / LPDDR-PIM



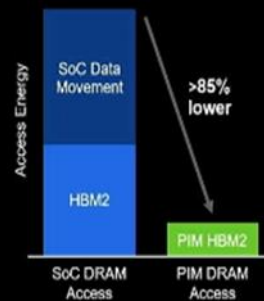
Energy Advantage of PIM on Generative AI

- Since **OpenAI** focuses on developing new AI technologies and pushing the boundaries of what can be done with AI, it is likely that they will explore the use of **PIM technology** in the future.
- In ISSCC 2023, AMD mentioned
 - Key algorithmic kernels can be **executed directly in memory**, saving precious communication energy
 - PIM can reduce energy by **85%** compared with conventional HBMs

Processing-in-Memory

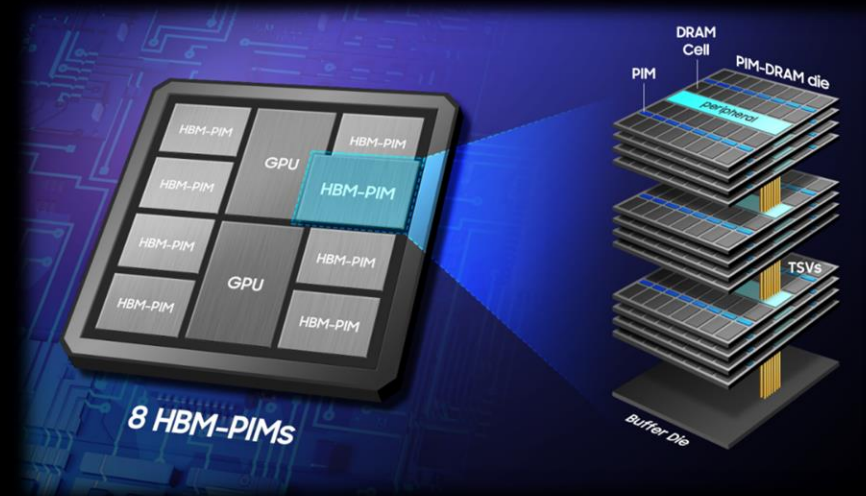


Key algorithmic kernels can be executed directly in memory, saving precious communication energy



[source] AMD ISSCC

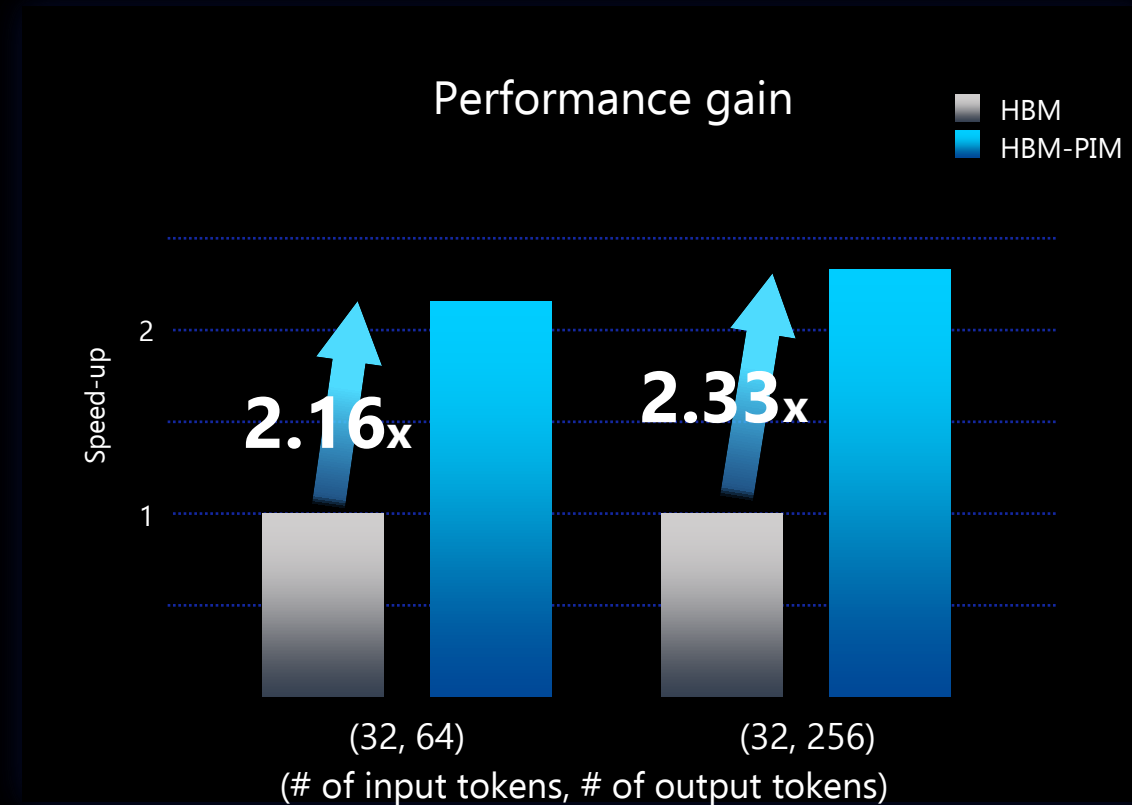
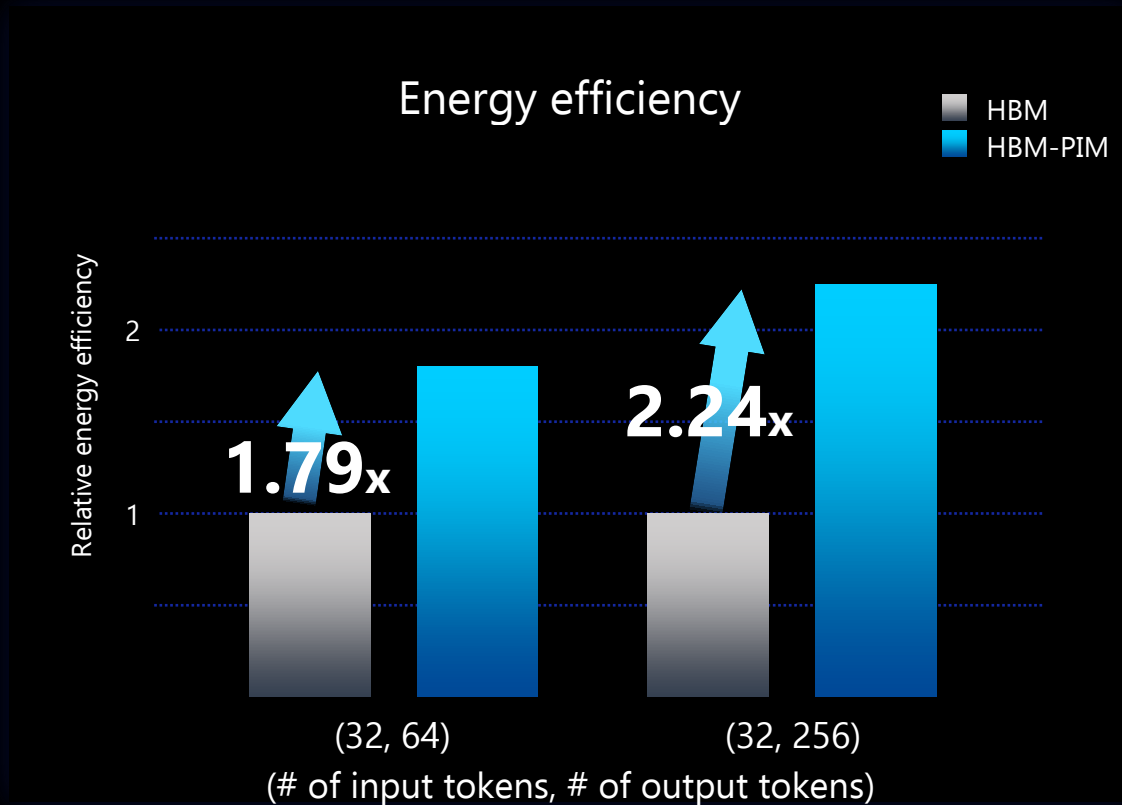
Future HBM-PIM concept



[source] Samsung, MemCon

Generative AI on HBM-PIM

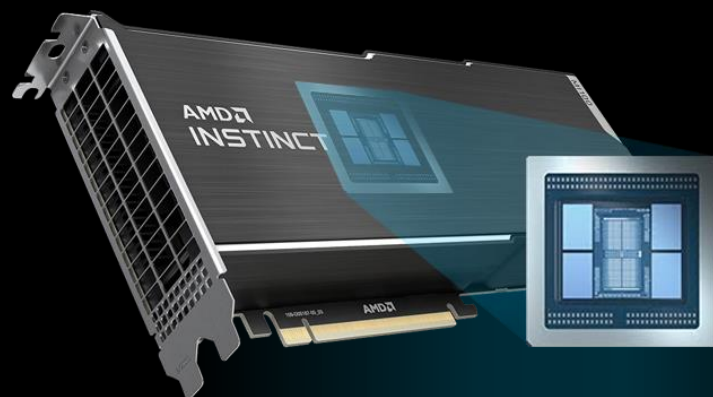
- Experimental setup: GPT-J (6B, 32 input tokens), single AMD MI100-PIM GPU
- About **2x** greater system energy efficiency compared to the GPU with a normal HBM
- GPT can be accelerated by more than **2x** over baseline



Architecture of HBM-PIM Cluster

- Installed 96 AMD MI100 GPUs fabricated with HBM-PIM
- Accelerate large-scale workloads with high energy efficiency and low latency

AMD MI100-PIM GPU



- Capacity : 24GB (4 cubes)
- PIM performance : 4.9 TFLOPS
- GPU performance : 184.6 TFLOPS (FP16)

Server node



- 8 MI100-PIM GPUs per node

HBM-PIM cluster



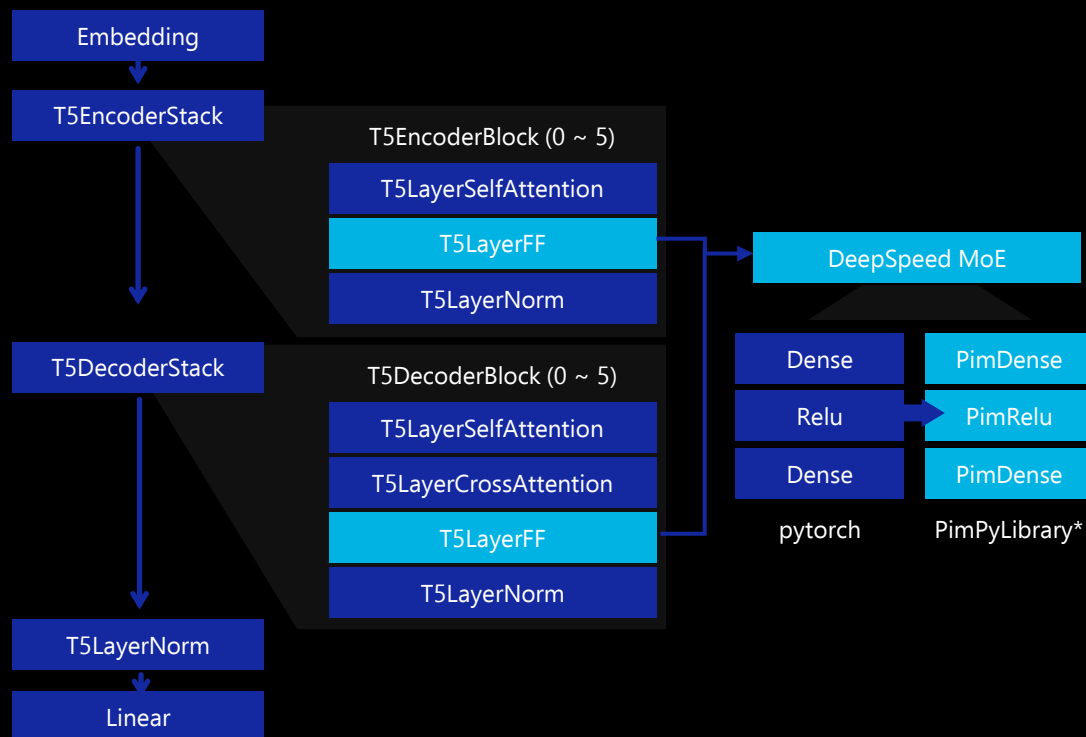
- Total 96 MI100-PIM GPUs in a cluster
- 12 nodes interconnected through 200G InfiniBand network (bi-section bandwidth : 1.2TB/s)
- Total memory capacity : 2.25TB
- Total PIM performance : 471.9 TFLOPS
- Total GPU performance : 17.7 PFLOPS (FP16)

Workload : T5(Transformer)-Based MoE Model

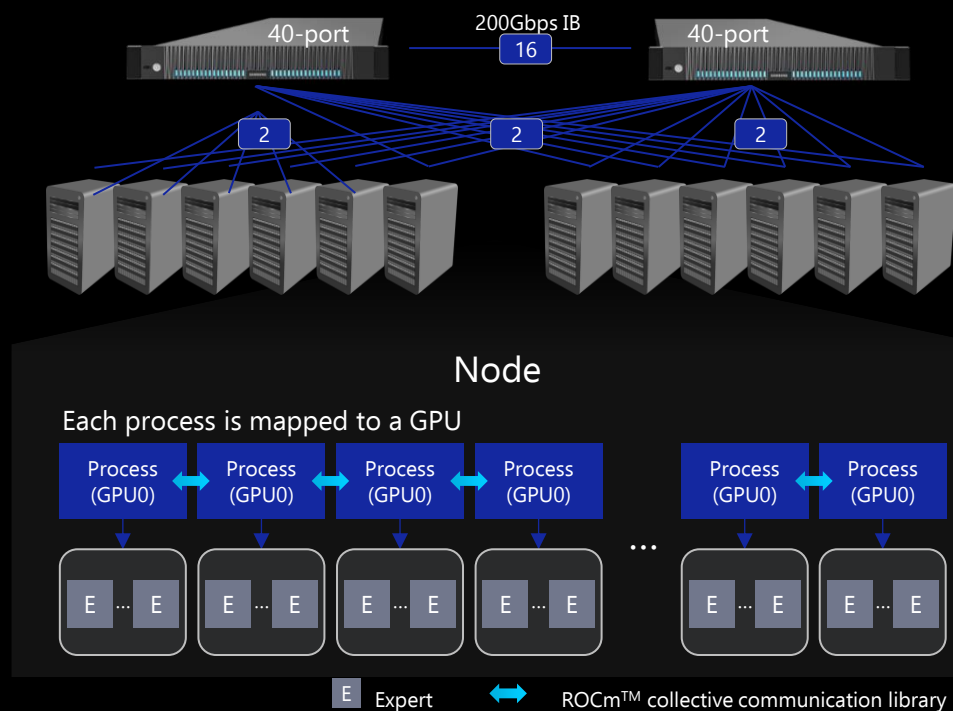
(Mixture of Experts)

- DeepSpeed MoE replaces “T5LayerFF” layers to accelerate T5-large model with PIM
- The MoE layers are updated to use our PimPyLibrary* APIs

T5-MoE model architecture



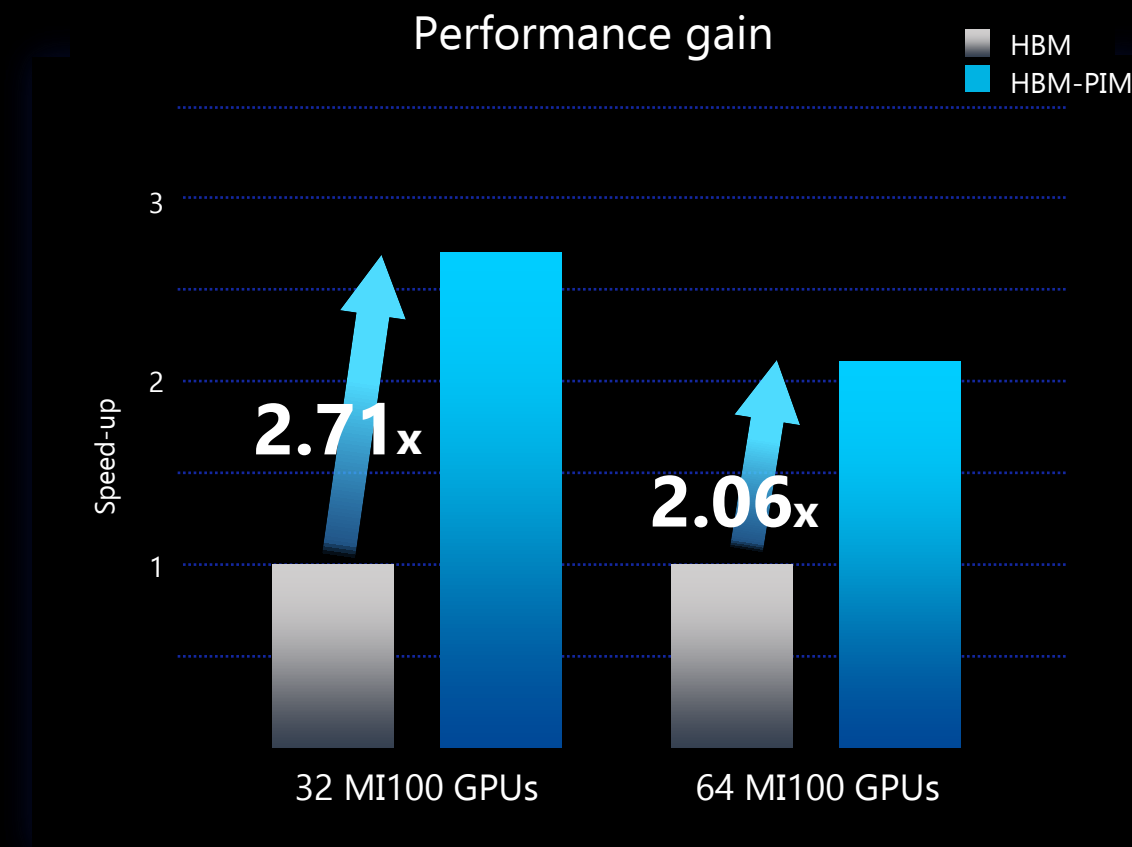
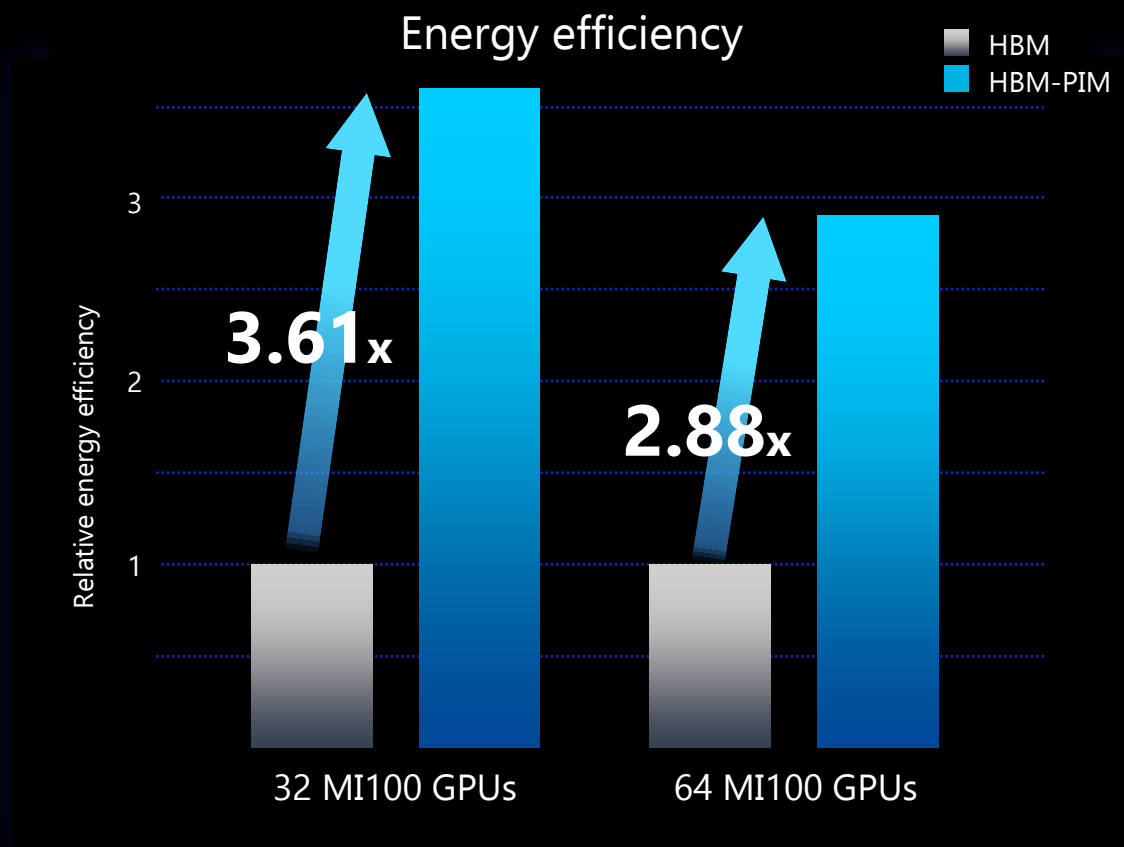
MoE deployment on HBM-PIM cluster



* PimPyLibrary is a python library for providing PIM-enabled AI operators. PIM SDK provides not only PimPyLibrary but also full SW stack for utilizing PIM

Energy Efficiency and Performance on MoE Model

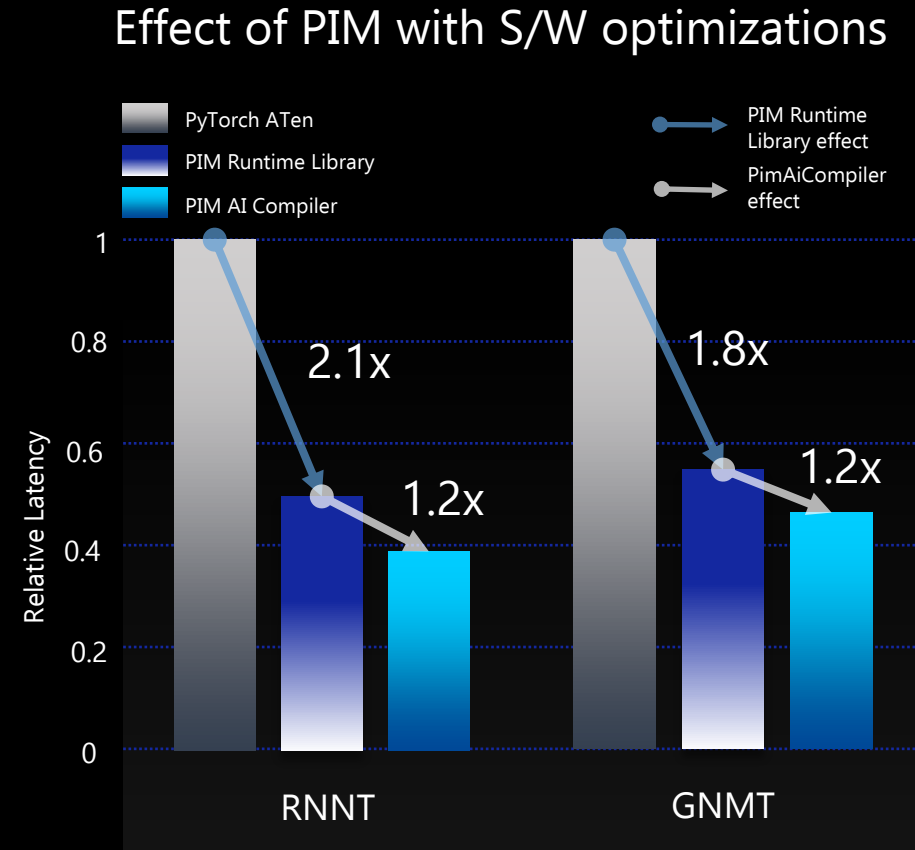
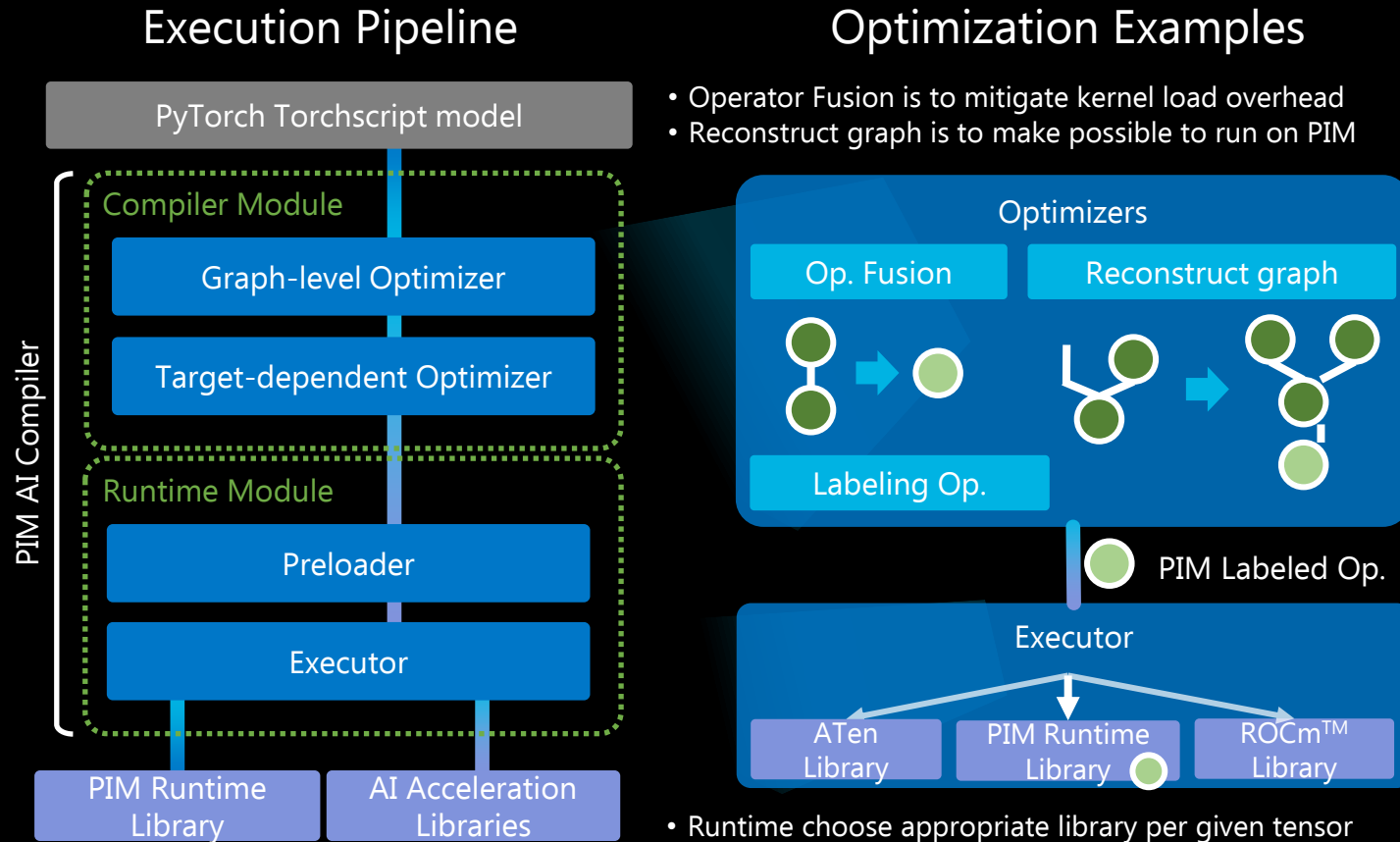
- More than **3x** greater system energy efficiency compared to normal GPU clusters
- Increases performance by more than **2x** over baseline



* Acknowledgement: Jaeyoung Heo and professor Sungjoo Yoo (Seoul National University) provided the idea of PIM acceleration for this workload

PIM S/W Stack for AI

- Support **existing AI frameworks** (e.g., PyTorch and TF) for users to utilize PIM functions
- **PIM Runtime Library**: Apply PIM and provide operator-level optimizations during PIM operation
- **PIM AI Compiler**: Provide graph-level optimizations during end-to-end execution



Promising Standard Programming Models

- **PIM-SYCL** accelerates upcoming HPC/AI applications on heterogeneous platform
 - SYCL supports CPU/GPU/NPU/FPGA in modern C++ template
- **PIM-OpenACC** is under development for legacy scientific applications
 - OpenACC enables incremental parallelization from C/Fortran serial code



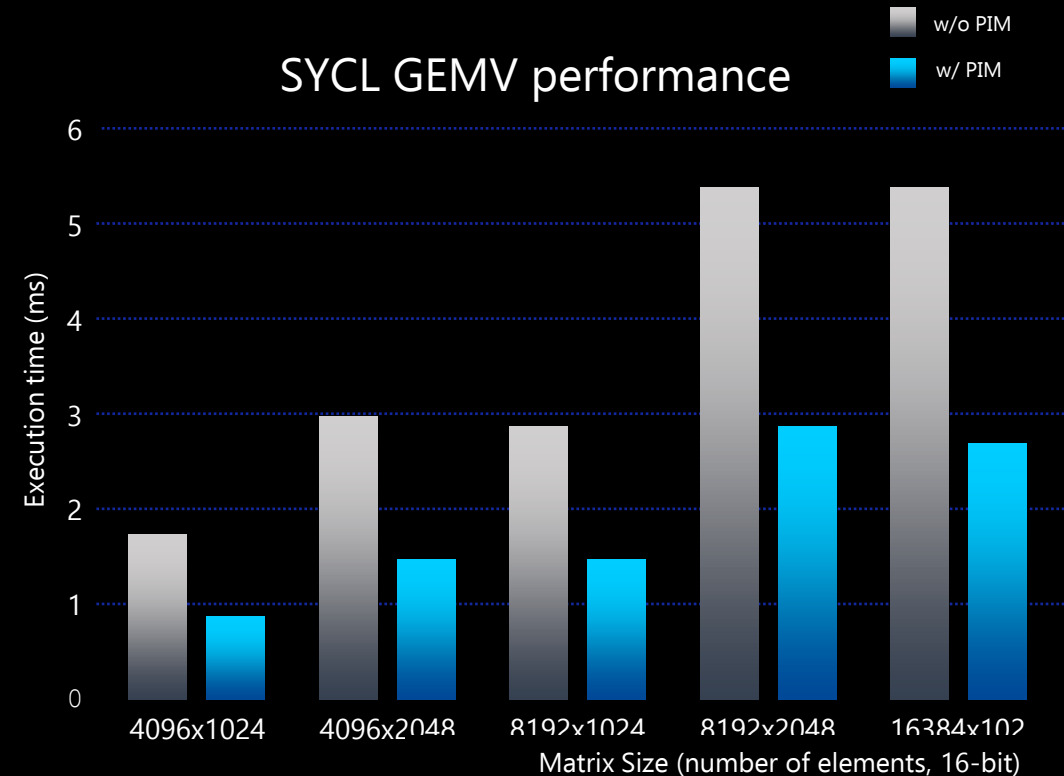
SYCL GEMV source code

```
// Buffer Allocation
buffer<sycl::half, 2> M{matrix, range<2>{N, N}};
buffer<sycl::half> X{in, range<1>{N}};
buffer<sycl::half> Y{out, range<1>{N}};

// Parallel Execution
q.submit([&](ext::samsung::pim_handler &ph) {
    ext::samsung::pim_accessor accM{M, ph, sycl::read_only};
    ext::samsung::pim_accessor accX{X, ph, sycl::read_only};
    ext::samsung::pim_accessor accY{Y, ph, sycl::write_only,
        sycl::property::no_init{}};

    ph.gemv(accY, accX, accM);
});
```

SYCL GEMV performance

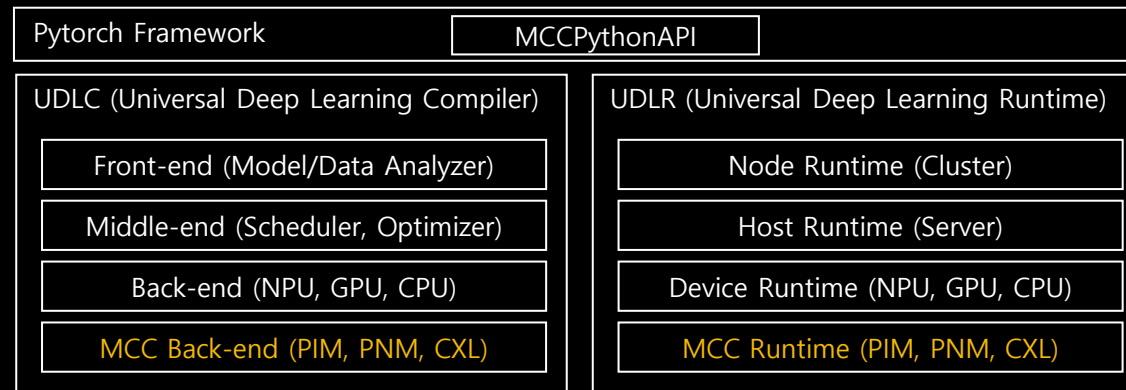


OneMCC S/W Standardization (To be)

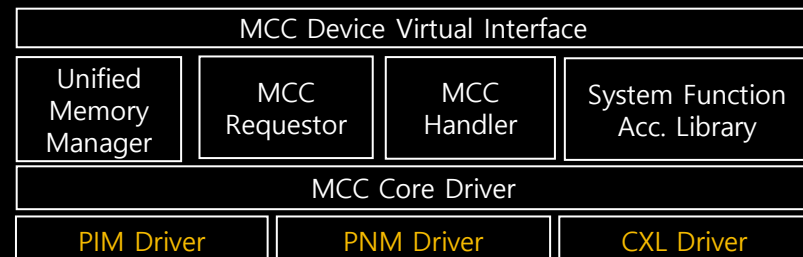
- OneMCC (Memory Coupled-Computing) is an open & standard S/W for **PIM, PNM, CXL solutions**
- Plan to provide standard programming model to support multi-architecture and domain
- Boost AI and HPC workloads with a variety of accelerators like CPUs, GPUs, and NPUs

OneMCC Infrastructure

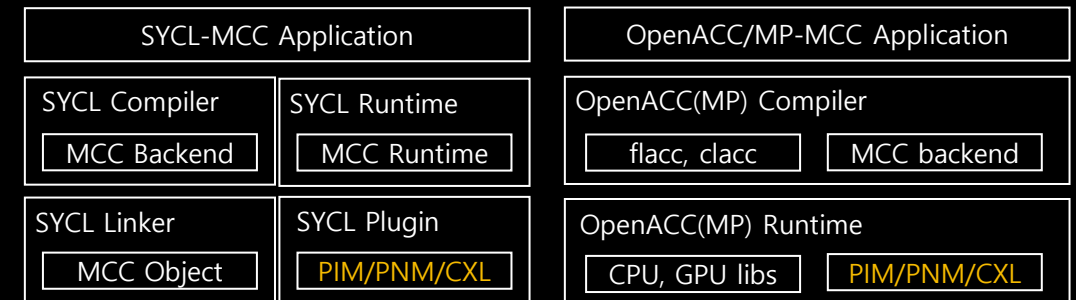
AI S/W Inference & Training



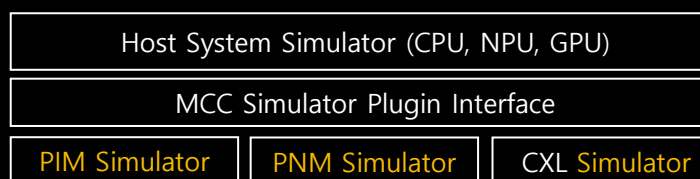
System S/W (Kernel Level Driver)



General Programming Model (SYCL, OpenACC/MP)



Simulation Infrastructure



Debug / Tools



Processing-in-Memory for On-device Generative AI

- Expanding On-device AI Necessity:
 - Data center **costs** and **power consumption** are increasing due to the growing demand for cloud AI
 - **Privacy** concerns are rising as sensitive data is transmitted to the cloud for processing
 - Network connectivity is not **always reliable** or available, particularly in remote areas
- **LPDDR-PIM** improves **battery life** by preventing memory over-provisioning just for bandwidth

Benefits of On-Device AI



Privacy Protection



Low Latency

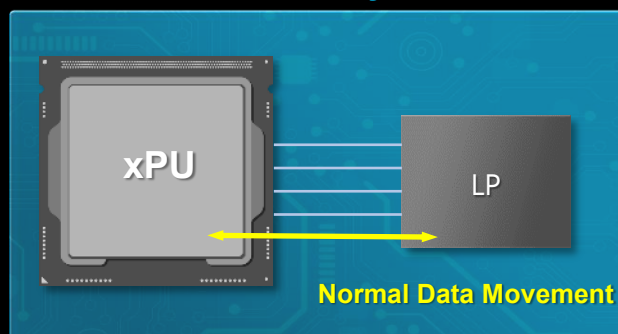


Low Power & Low Cost

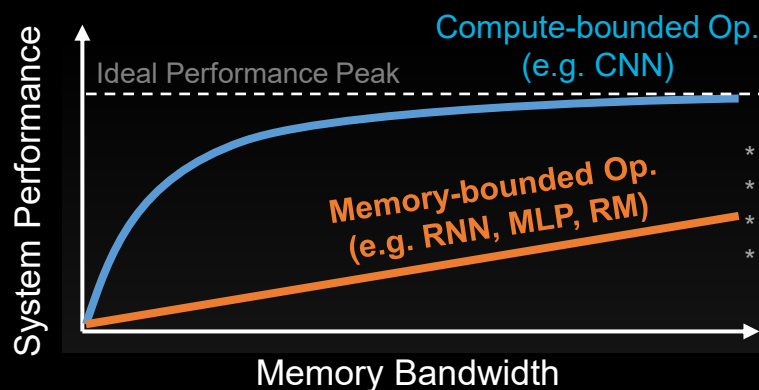
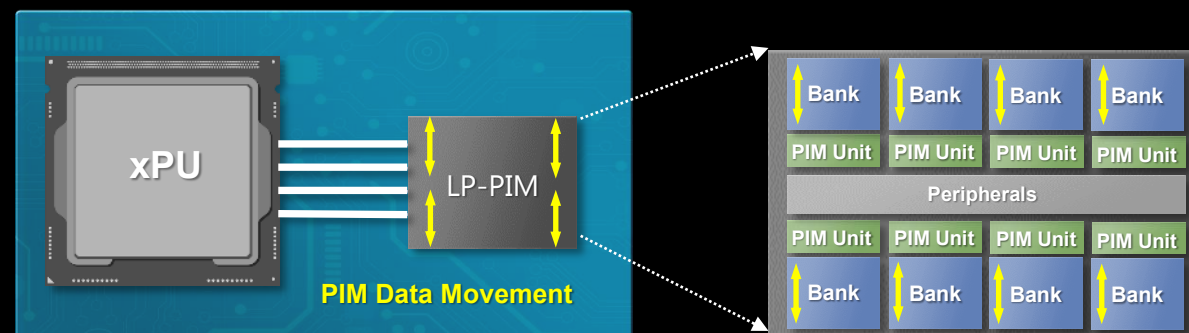
LPDDR-PIM Concept

- Improve the **4.5x** performance and save **72%** of energy in the system with in-DRAM processing
 - Performance: Utilize up to **8x** higher in-DRAM bandwidth by multi-bank parallel operation
 - Energy Efficiency: Reduce data movement energy by utilizing PIM unit

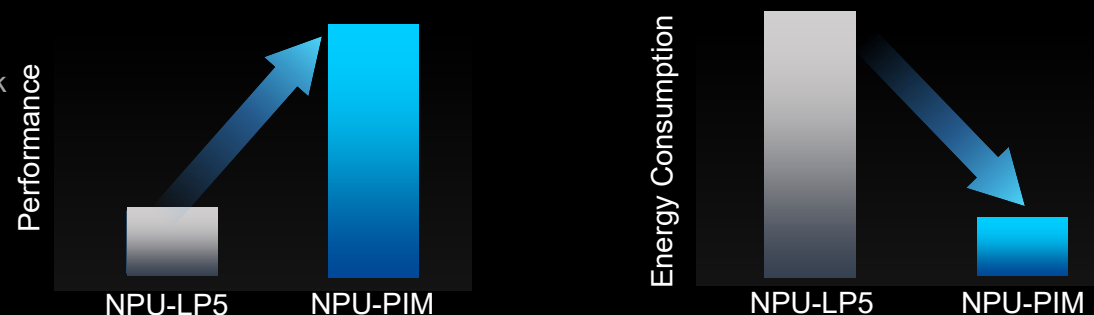
LP5-xPU System



PIM-xPU System



System Performance and Energy Comparison

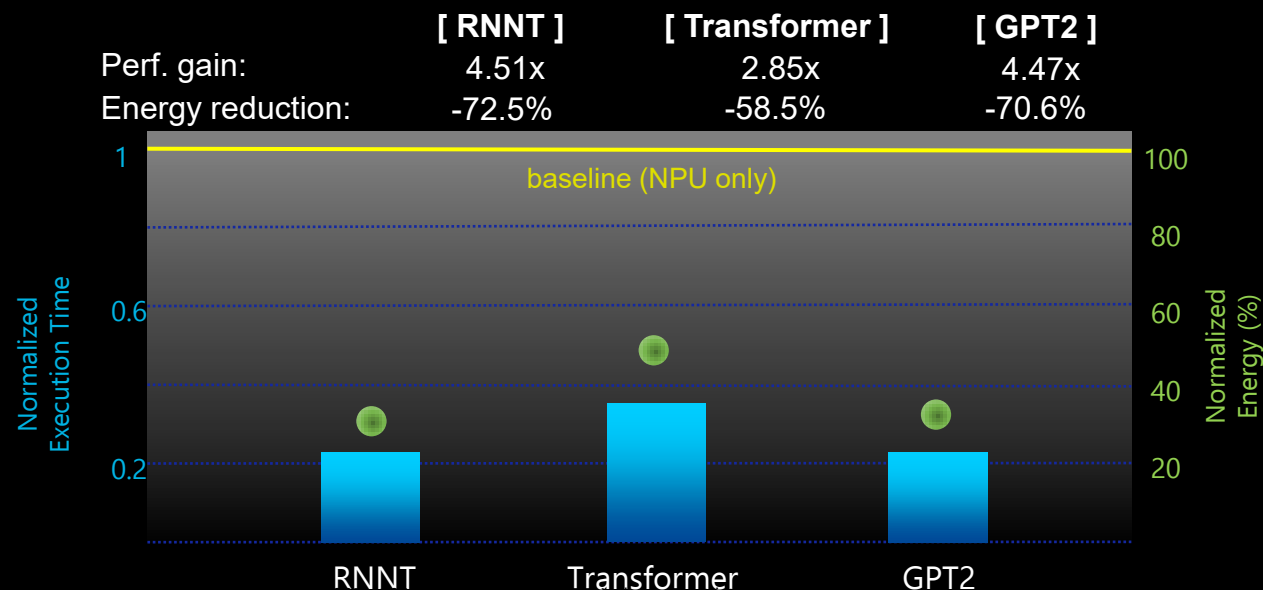
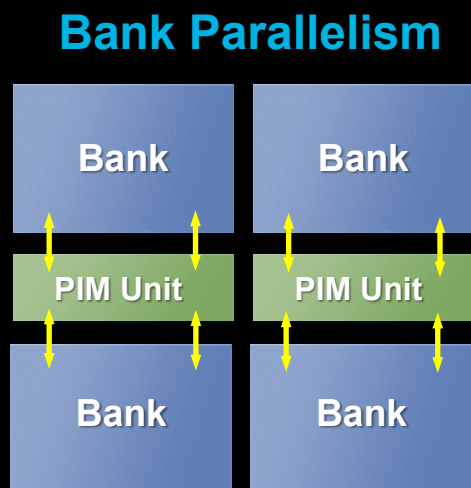


LPDDR-PIM Features

- Peak internal bandwidth: **102.4 GB/s**
 - Using Bank-level parallelism, 8x bandwidth of the base LPDDR product
- Supporting native integer/floating point arithmetic and logical (and/or/...) operations
- Peak performance: 102.4 GFLOPS/s (FP16), 204.8 GOPS/s (INT8)
- Acceleration target: memory-bounded operations such as **BLAS1 and BLAS2**
- Samsung can support LPDDR-PIM simulator package to measure performance gain & energy reduction

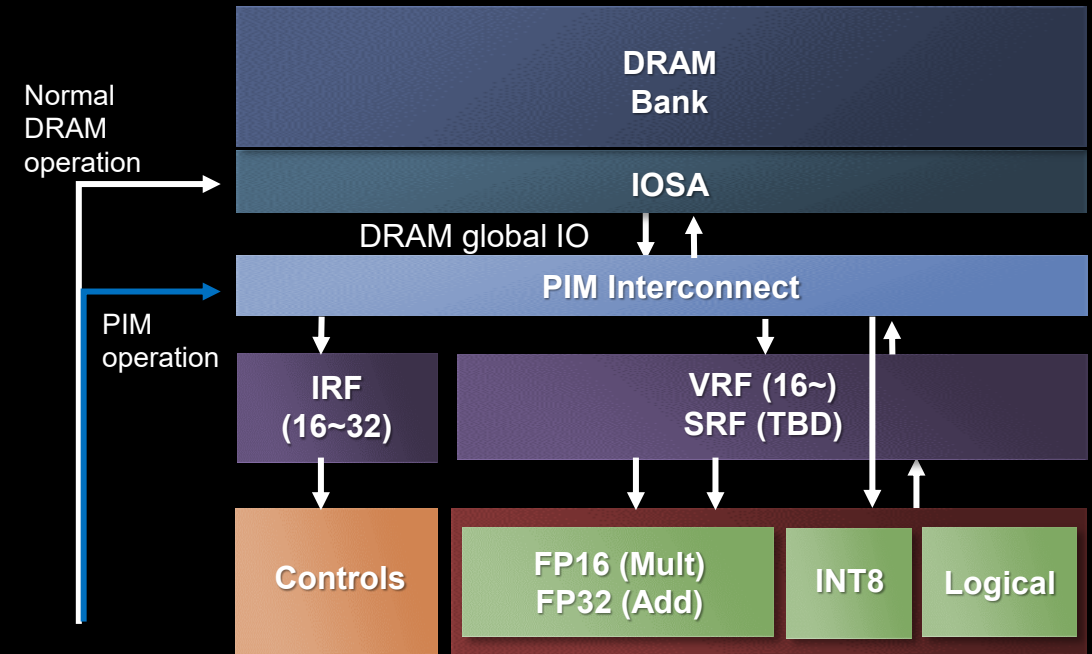
BLAS1: Element-wise addition/multiplication
or layer normalization
BLAS2: vector-matrix multiplications

End-to-end inference on AI application (simulation-based)



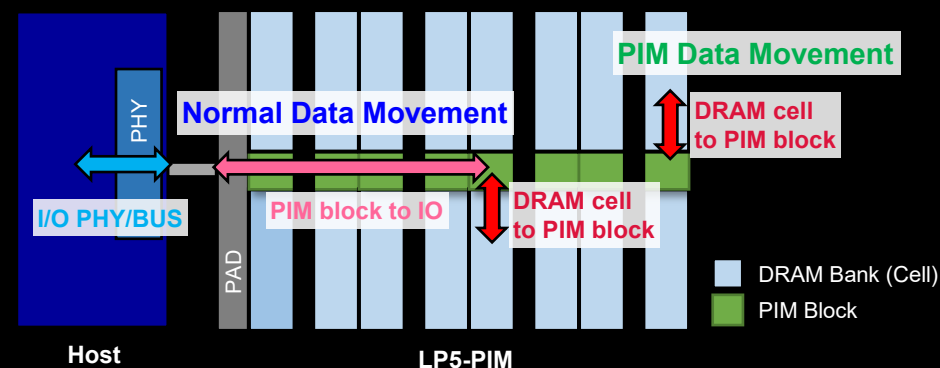
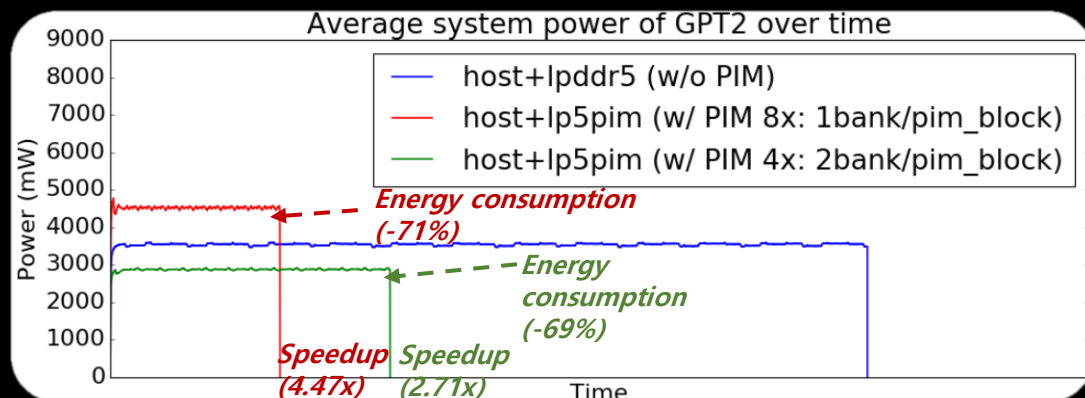
LPDDR-PIM Architecture

- PIM Unit is placed between every 1 bank (maximum performance) or 2 banks (moderate area overhead)
- PIM Unit: 256-bit SIMD FPU and registers (~640 bytes per PIM block)
 - Supporting operations: FP16 multiplication, FP32 accumulation, int8 arithmetic, etc.
 - PIM registers : Instruction (IRF), Vector (VRF), and Scalar (SRF)

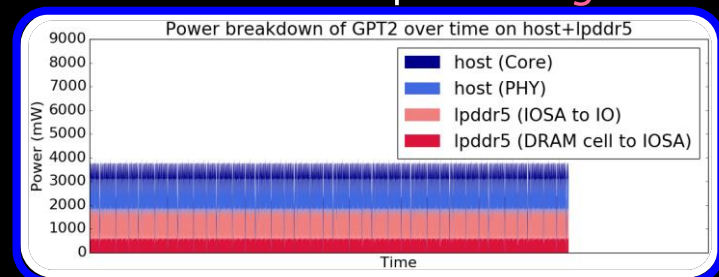


LPDDR-PIM System Performance/Power Analysis

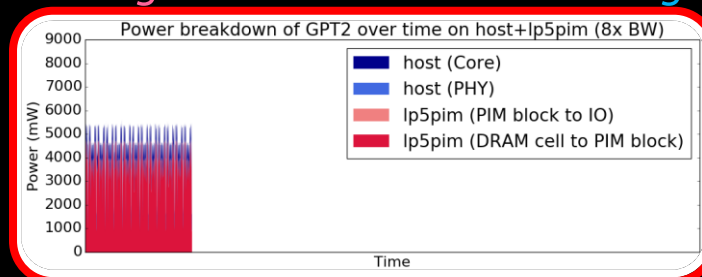
- Evaluate the performance and power consumption of GPT2
 - LP5-PIM improves energy efficiency by **shorter execution time**



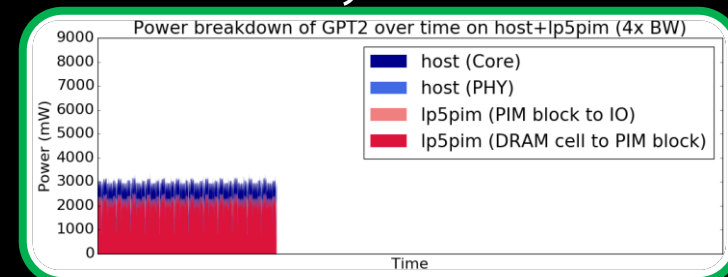
- Power consumption of **DRAM internal component (red)** increases proportionally
- Power consumption of **global I/O bus (light red)** and **I/O PHYs (light blue)** considerably decreases



[host + lpddr5]



[host + lp5pim (8x: 1 bank/PIM block)]



[host + lp5pim (4x: 2 bank/PIM block)]

* The power ratio is 17.9% (DRAM cell to IOSA), 31.3% (IOSA to IO), 31.1% (PHY), and 19.7% (core) for baseline.

* The power ratios are 85.3% (DRAM cell to PIM block), 14.6% (core), and 0.1% (etc) for lp5-pim(8x), and 76.9% (DRAM cell to PIM block), 23% (core), and 0.1% (etc) for lp5-pim(4x), respectively.

* Simulation experiment uses 4 memory (lp5/pim) channels

* lp5pim (DRAM cell to PIM block) includes ACT, PRE, IDLE, and REF

* The result of simulation is that more than 99% of memory traffic decreases by PIM

* host (core) includes processing and IDLE power

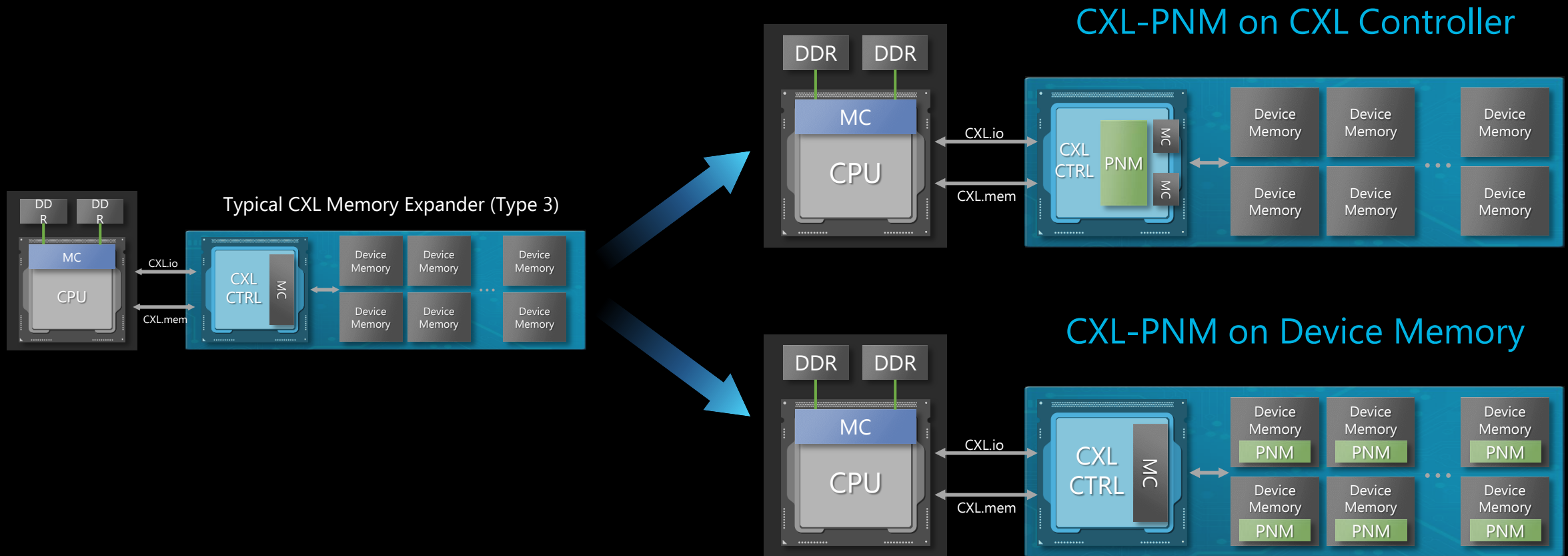


CXL-PNM

Industry's 1st CXL-PNM (Processing-near-Memory)

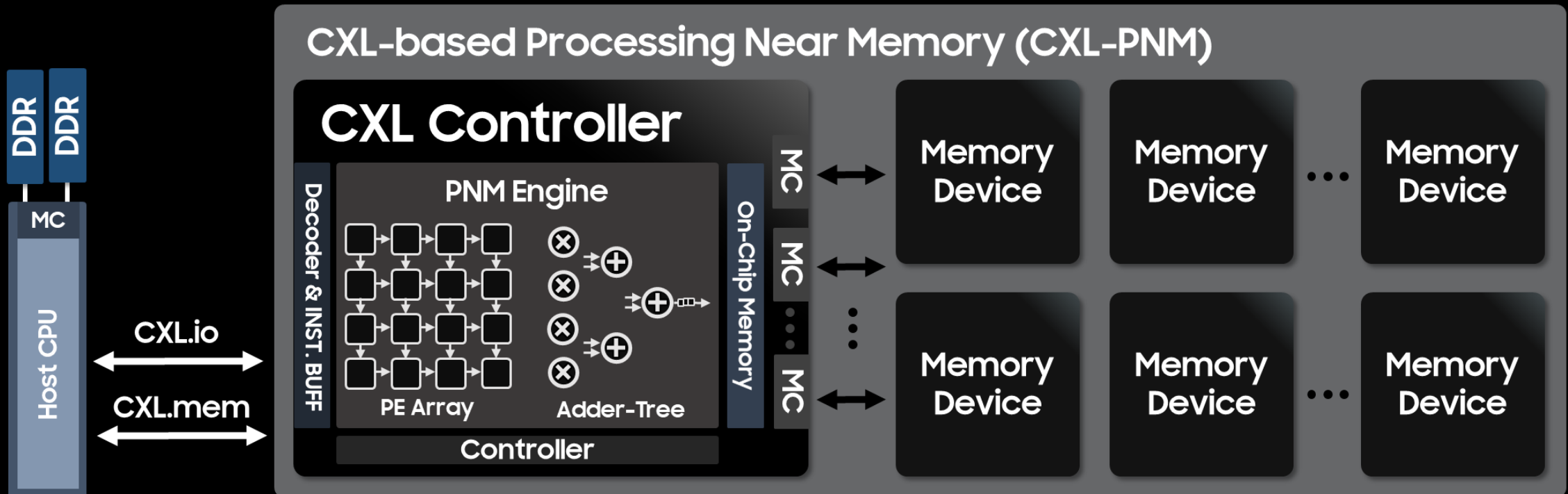
CXL-PNM Architecture

- A CXL-based Processing-near-Memory (PNM) Solution
- Two types of CXL-PNM: on CXL controller and on device memory



CXL-PNM Architecture for GPT

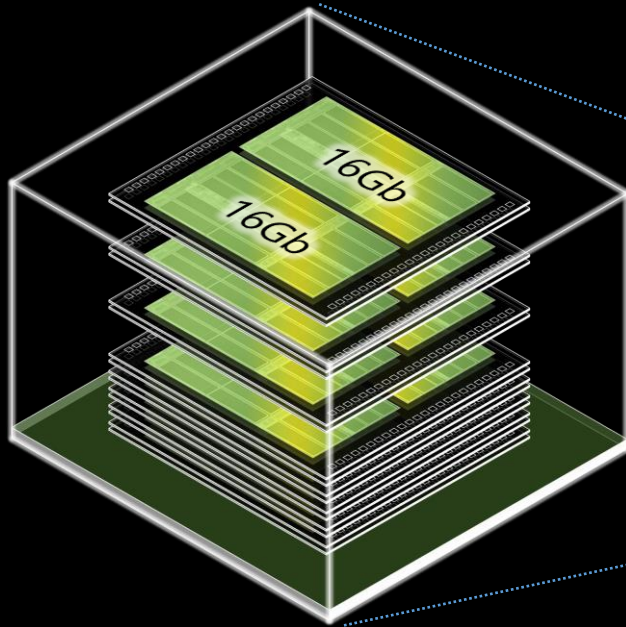
- Heterogeneous compute unit (PE Array and Adder-Tree) on PNM Engine
 - Adder-tree designed to perform GEMV operation (Generation stage)
 - PE Array for acceleration of GEMM operation (Summarization stage)



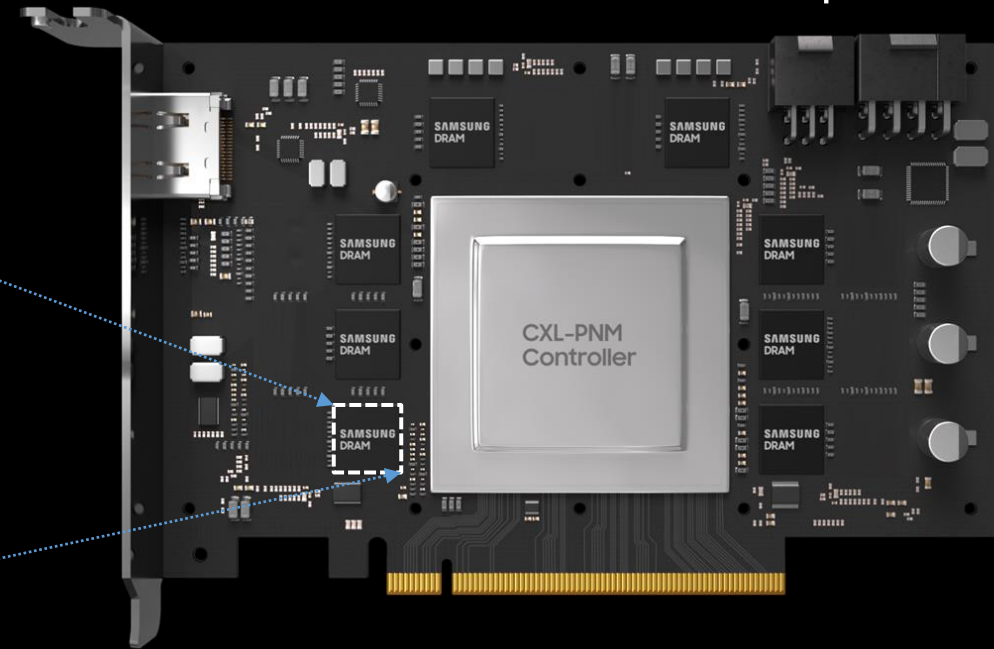
512GB 1.1TB/s CXL-PNM Concept

- CXL-PNM is able to be used for a wider range of systems including AI/ML accelerators
- Compared to other solutions, it can give unique trade-off among **capacity, bandwidth, and power**
- CXL-PNM can provide **512GB capacity** and **1.1TB/s bandwidth**

512Gb DRAM Package

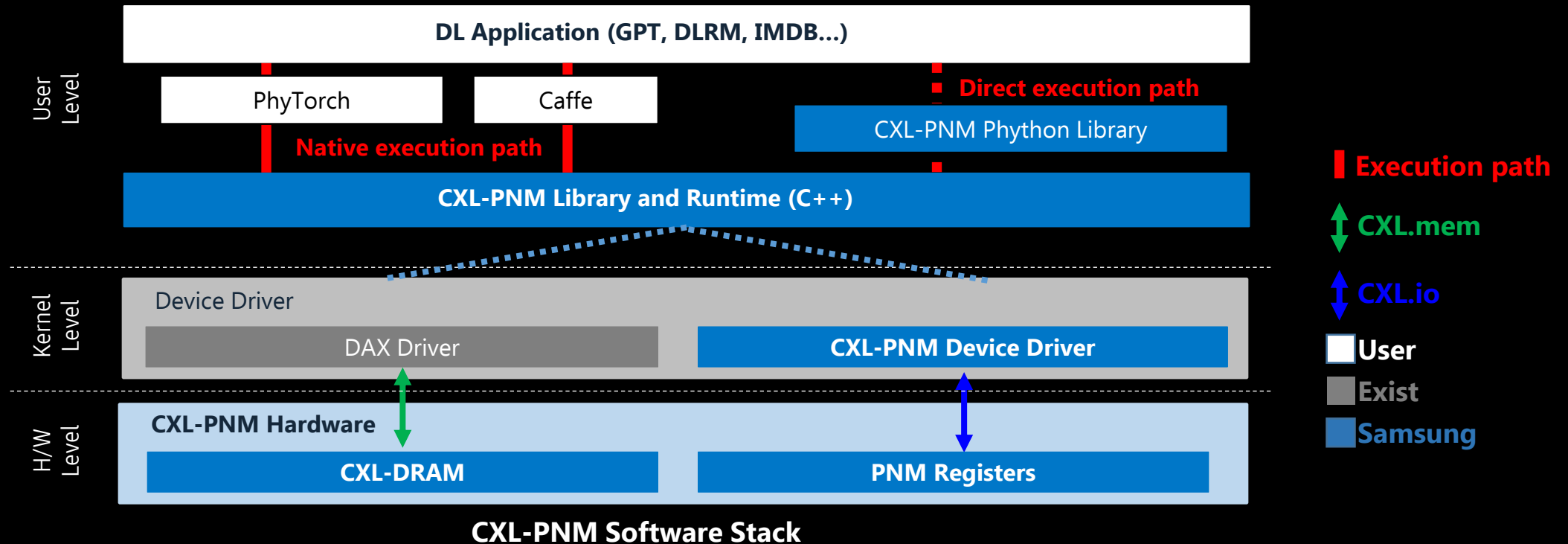


CXL-PNM Module Concept



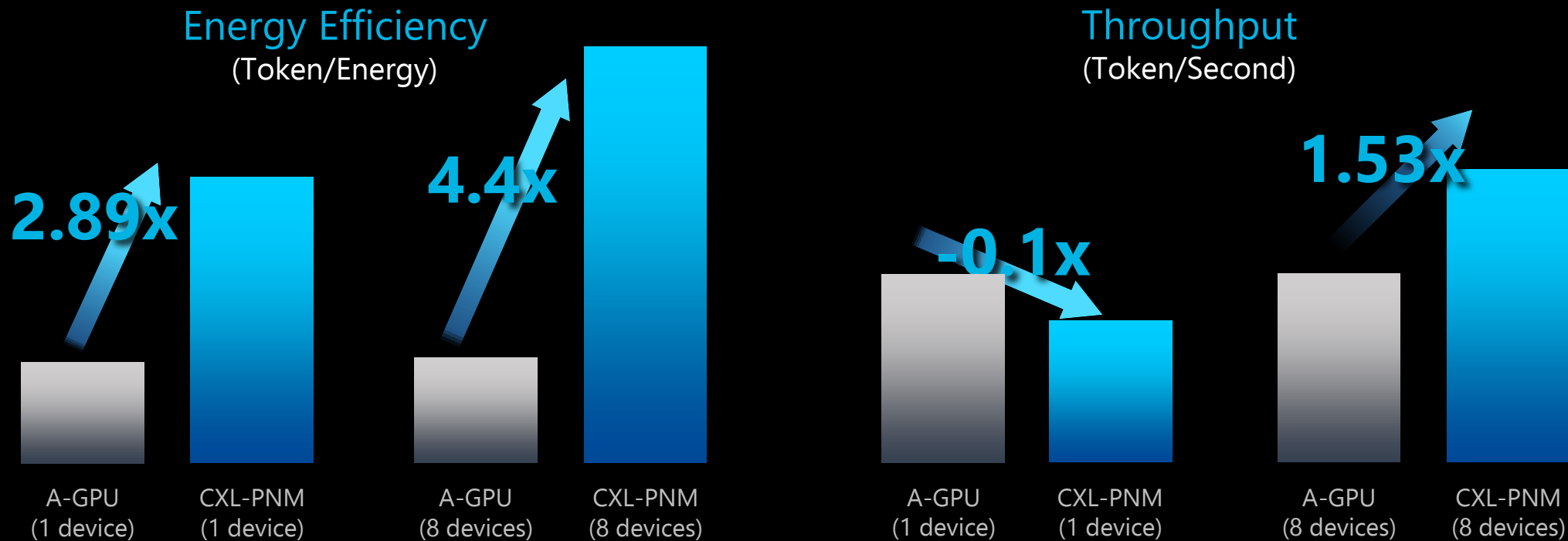
CXL-PNM Software Stack

- CXL-PNM software stack for users to seamlessly and transparently utilizes the CXL-PNM platform
- CXL-PNM software stack includes user-level of library, runtime and kernel level of device driver
- CXL-PNM software stack supports two execution paths
 - Native execution path – Automatically offload PNM operation without modification of application source code
 - Direct execution path – Explicitly call PNM operations on the user application



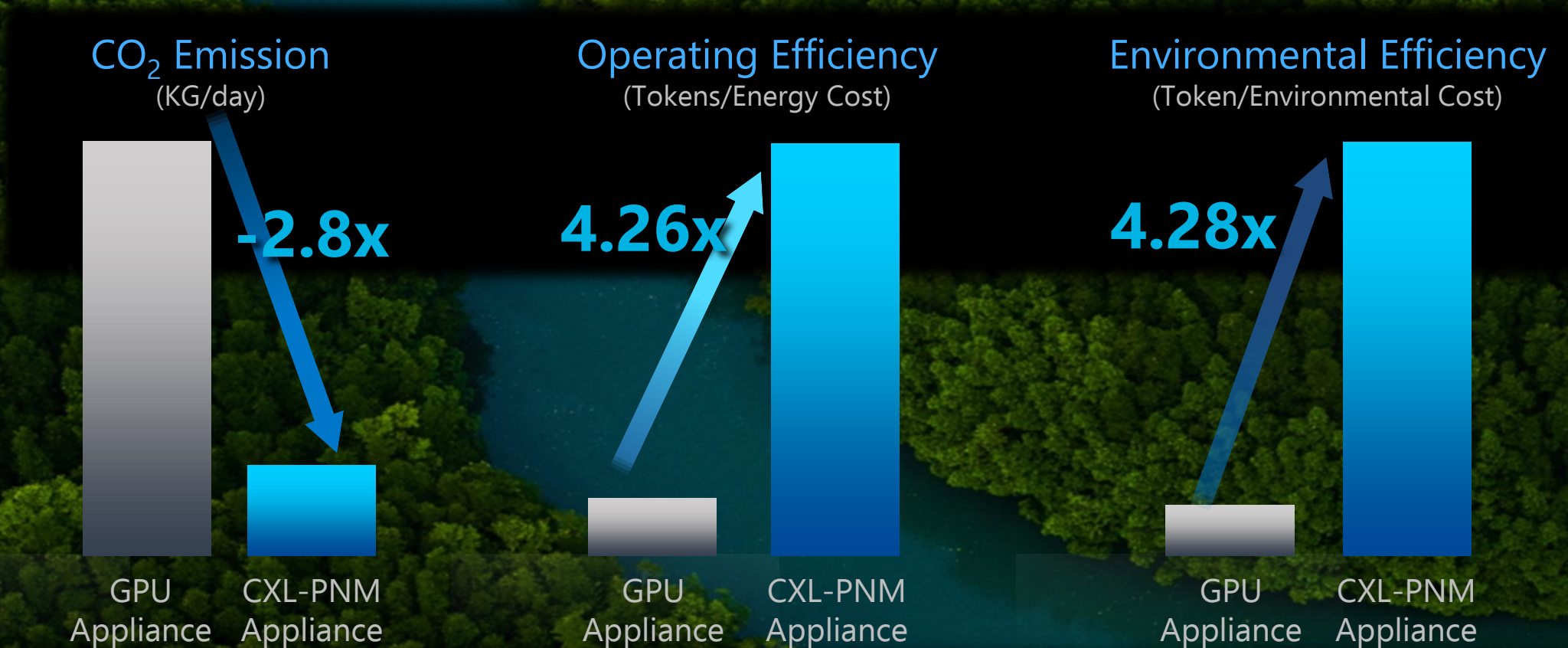
Energy and Throughput Comparison for LLM

- End-to-end performance evaluation of GPU and CXL-PNM with the same number of devices
 - CXL-PNM gives **2.9×** higher energy efficiency, only **10.8%** lower throughput, compared to A-GPU
 - However, CXL-PNM with large capacity can accelerate large-scale LLMs w/o any communication overhead
 - Multiple CXL-PNM can offer **4.4×** higher energy efficiency and **53%** higher throughput than multiple GPUs



CO₂ Emission Reduction by CXL-PNM

- Operating/Environmental (Energy/CO₂) cost of GPU/CXL-PNM appliance with eight devices
 - GPU appliance is **2.8×** more expensive than CXL-PNM appliance for the environmental cost
 - Operating efficiency of CXL-PNM appliance reduces the amount of CO₂ emission
 - CXL-PNM appliance is **4.3×** more efficient than that of GPU appliance



Samsung AI memory